



Lakshya JEE 2.0 (2024)

Method of Differentiation

DPP-01

1. If $y = \sqrt{x} + x\sqrt{x} + x^2\sqrt{x} + x^3\sqrt{x}$, find $\frac{dy}{dx}$

(1) $\frac{1}{2\sqrt{x}} + \frac{3}{2}\sqrt{x} + \frac{5}{2}x^{1/2} + \frac{7}{2}x^{5/2}$

(2) $\frac{1}{2\sqrt{x}} + \frac{3}{2\sqrt{x}} + \frac{5}{2}x^{3/2} + \frac{7}{2}x^{5/2}$

(3) $\frac{1}{2\sqrt{x}} + \frac{3}{2}\sqrt{x} + \frac{5}{2}x^{3/2} + \frac{7}{2}x^{5/2}$

(4) $\frac{1}{2\sqrt{x}} + \frac{3}{2}\sqrt{x} + \frac{5}{2}x^{1/2} + \frac{7}{2}x^{3/2}$

2. If $y = \log(\sin(3x+5))$, find $\frac{dy}{dx}$

(1) $3\cot(5x+3)$ (2) $3\cot(3x+5)$

(3) $3\tan(3x+5)$ (4) $3\tan(5x+3)$

3. If $y = \log(\sqrt{x-1} - \sqrt{x+1})$, find $\frac{dy}{dx}$

(1) $\frac{-1}{2\sqrt{x-2}}$ (2) $\frac{1}{2\sqrt{x^2-2}}$

(3) $\frac{-1}{2\sqrt{x^2-1}}$ (4) $\frac{1}{2\sqrt{x^2+2}}$

4. If $y = \log(\sin x + \cos x)$, find $\frac{dy}{dx}$

(1) $\cot\left(\frac{\pi}{4} - x\right)$ (2) $\cot\left(\frac{\pi}{4} + x\right)$

(3) $\tan\left(\frac{\pi}{4} + x\right)$ (4) $\tan\left(\frac{\pi}{4} - x\right)$

5. Let f be a function for which $f'(x) = \frac{1}{x^2+1}$,

If $g(x) = f(3x-1)$, find $g'(x)$.

(1) $\frac{3}{9x^2-6x+2}$ (2) $\frac{3}{9x^2+6x-2}$

(3) $\frac{3}{3x^2+2x-2}$ (4) $\frac{3}{3x^2-2x+2}$

6. If $y = \log_{\sin x}(\tan x)$, then $\left(\frac{dy}{dx}\right)_{\pi/4}$ is equal to:

(1) $\frac{4}{\log 2}$ (2) $-4 \log 2$

(3) $\frac{-4}{\log 2}$ (4) None of these

7. If $y = \sqrt{\log x + \sqrt{\log x + \sqrt{\log x + \dots \infty}}}$, then $\frac{dy}{dx}$ is:

(1) $\frac{x}{2y-1}$ (2) $\frac{x}{2y+1}$

(3) $\frac{1}{x(2y-1)}$ (4) $\frac{1}{x(1-2y)}$

8. If $f(x) = \frac{x-1}{x+1}$, find $\frac{d(f(f(f(f(x))))}{dx}$

(1) $\frac{2}{(x-1)^2}$ (2) $\frac{2}{(x-1)^3}$

(3) $\frac{-2}{(x-1)^2}$ (4) $\frac{-2}{(x-1)^3}$

9. If $f(x) = \log_x(\ln x)$, then $f'(x)$ at $x = e$ is _____.

10. If $y = \cos^{-1}(\cos x)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{4}$, then:

(1) 1 (2) -1

(3) $\frac{1}{\sqrt{2}}$ (4) None of these

11. Let $f(x) = \begin{vmatrix} 1 & 2x & 3x^2 \\ x & x^2 & x^3 \\ 0 & 2 & 6x \end{vmatrix}$

Then find the value of $f'(1)$

(1) 0 (2) -2

(3) -6 (4) 6



12. Let $f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2\cos 2x \\ \cos 3x & \sin 3x & 2\cos 3x \end{vmatrix}$. Then find

the value of $f'\left(\frac{\pi}{2}\right)$

(1) 0

(2) 1

(3) 2

(4) 3



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (3)
2. (2)
3. (3)
4. (4)
5. (1)
6. (3)

7. (3)
8. (1)
9. (1/e)
10. (2)
11. (3)
12. (2)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>